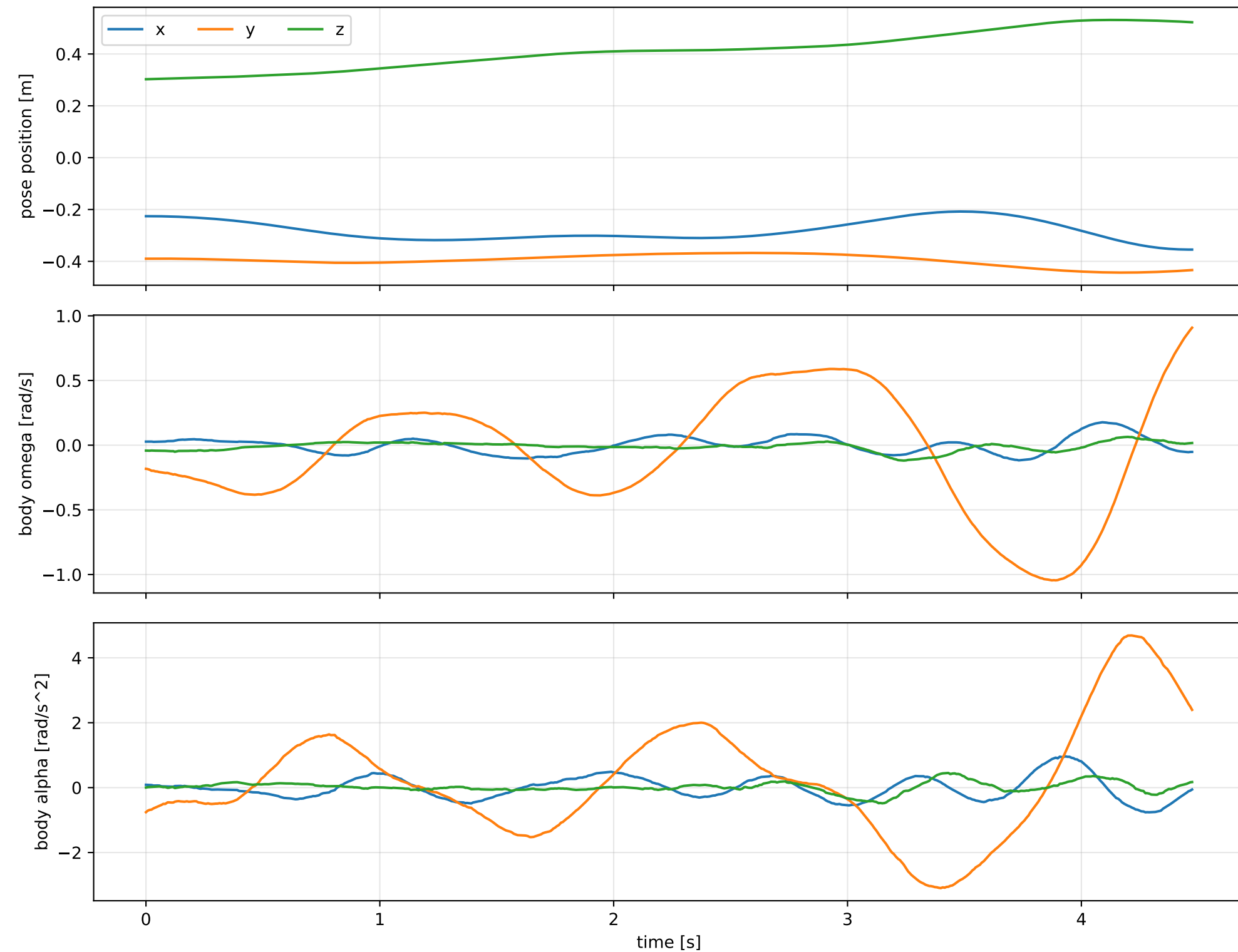
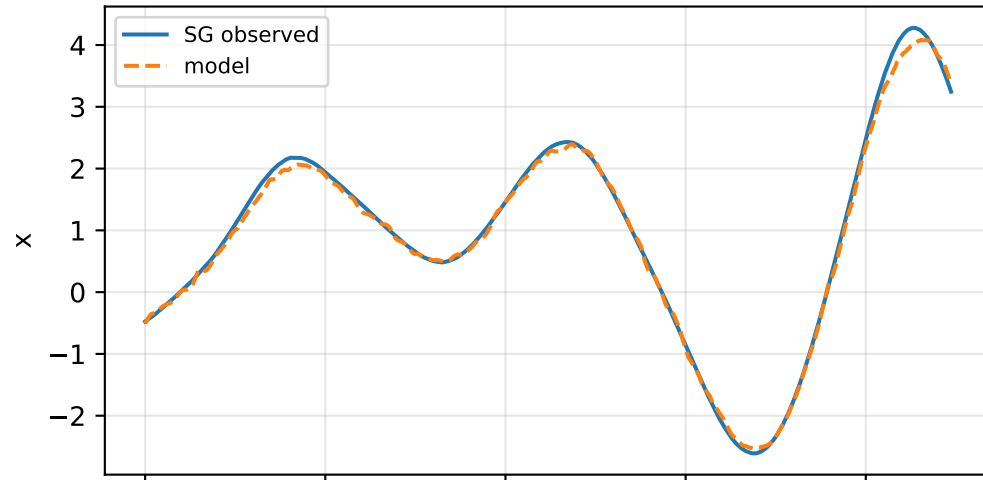


gimbal_measured_linear: SG trajectory and derived motion

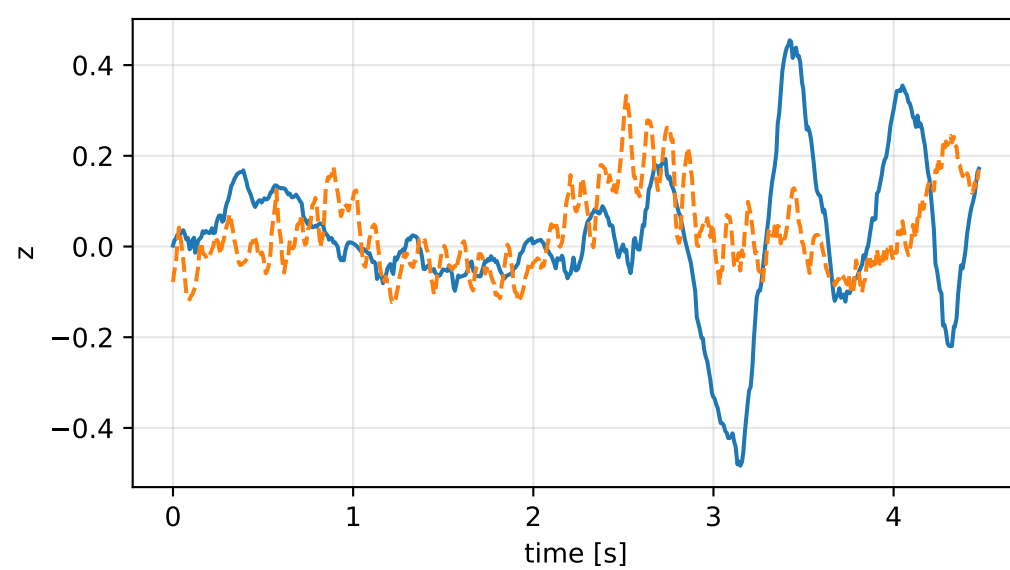
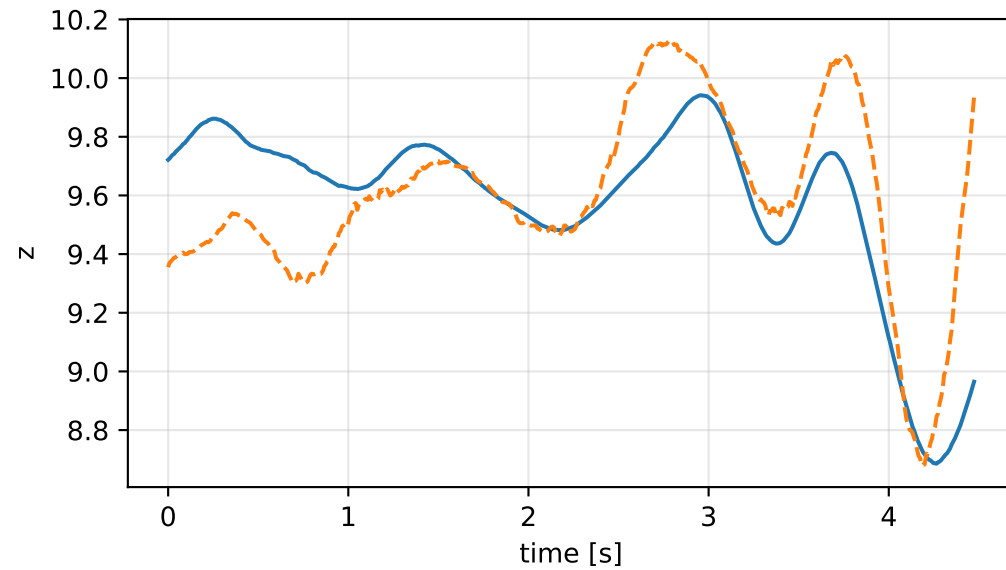
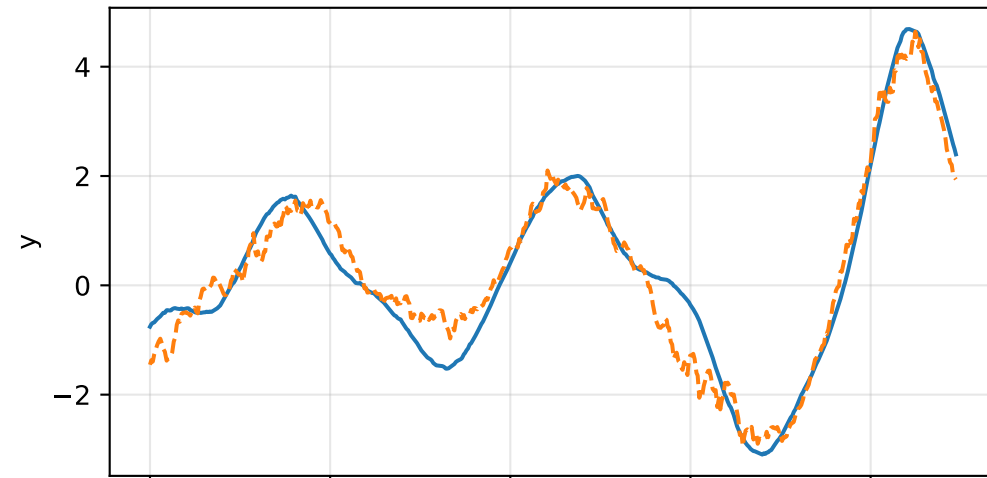
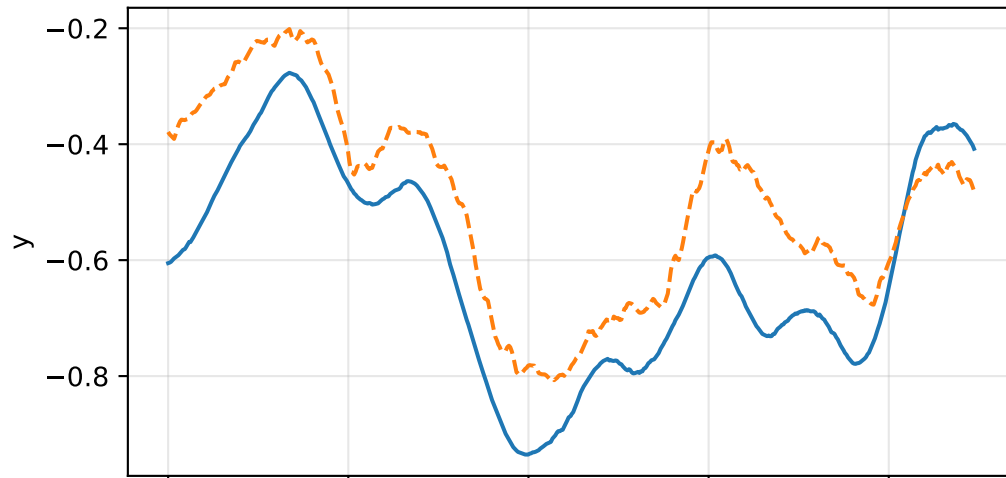
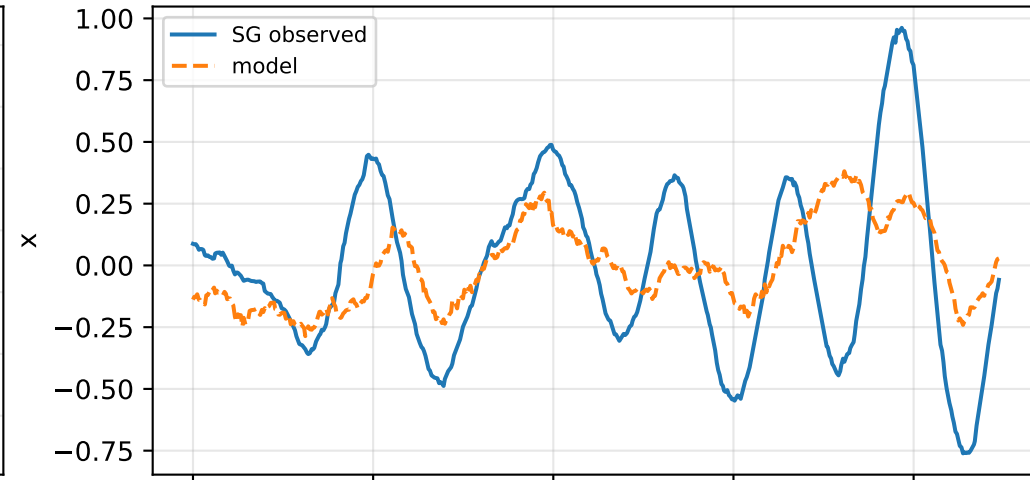


gimbal_measured_linear: acceleration residual objective

specific acceleration [m/s^2]

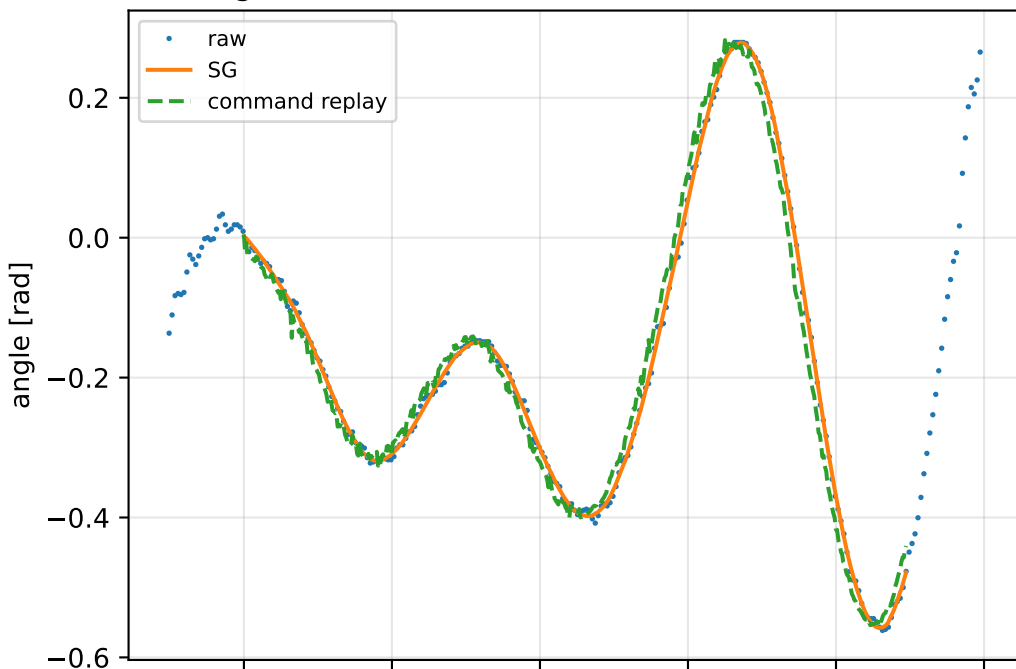


angular acceleration [rad/s^2]

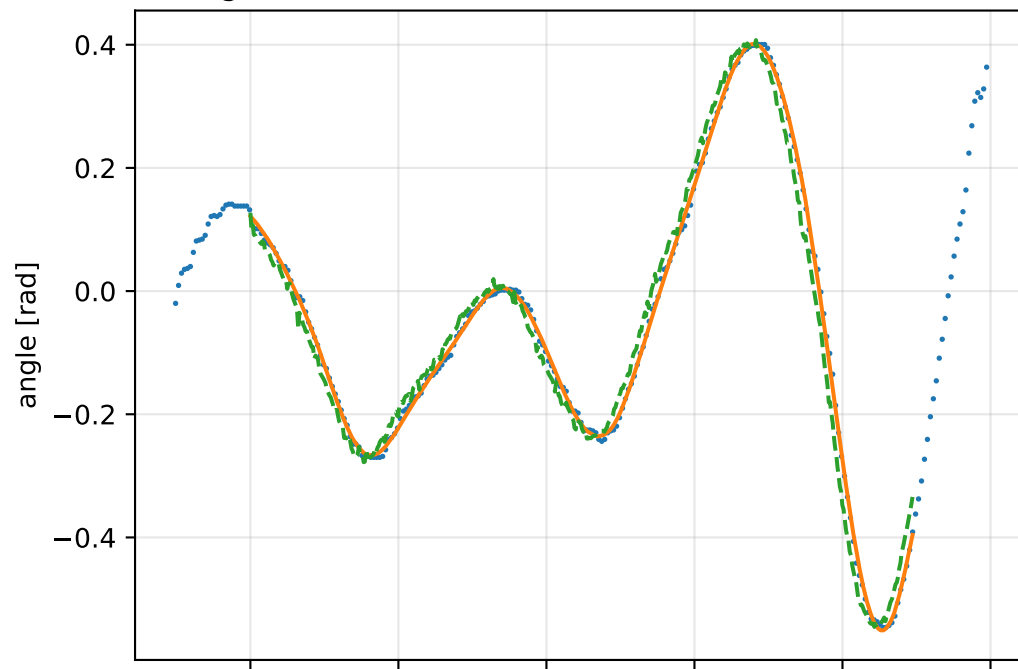


gimbal_measured_linear: gimbal measurement audit (objective=measured_linear)

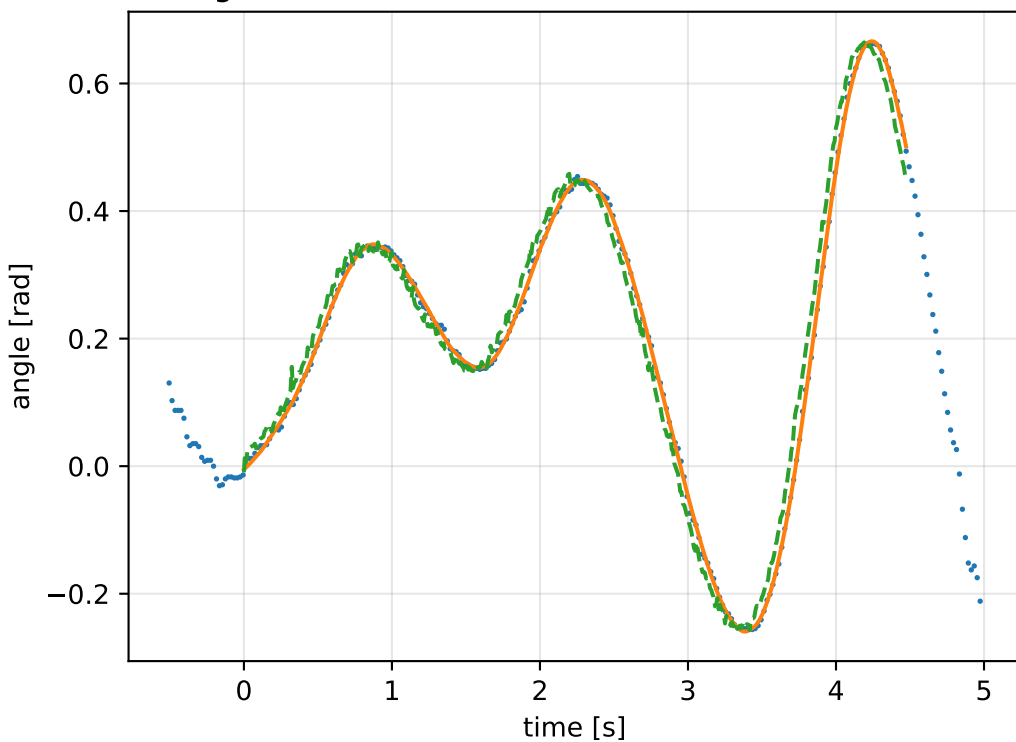
gimbal 1: RMSE=0.027, max=0.07529 rad



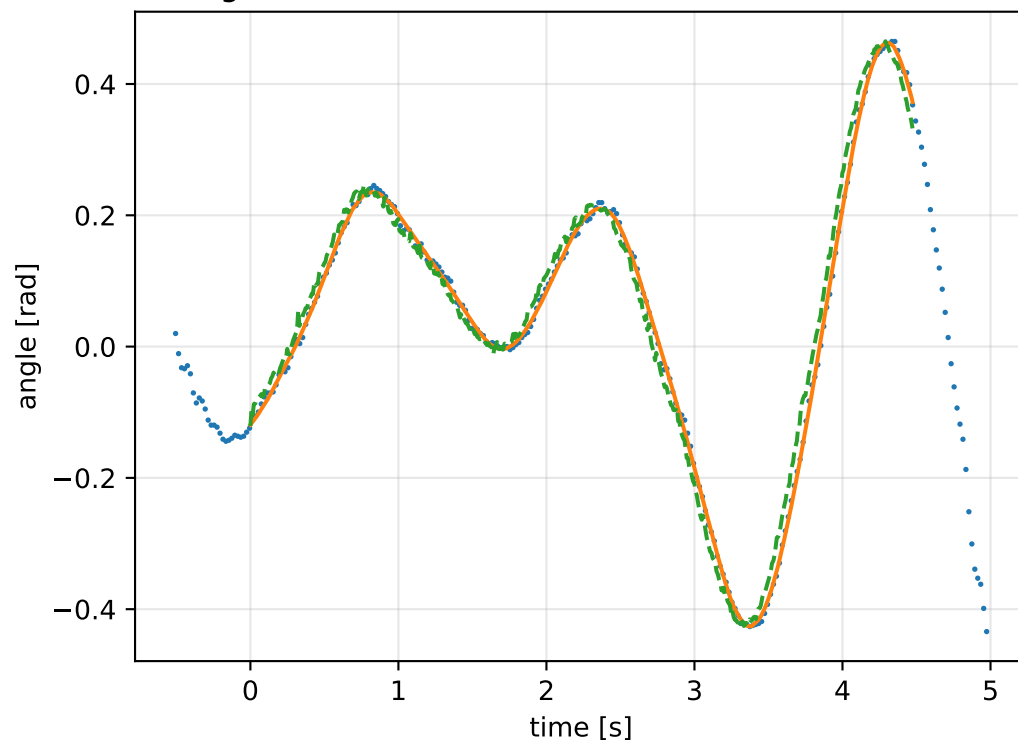
gimbal 2: RMSE=0.03107, max=0.08239 rad



gimbal 3: RMSE=0.03106, max=0.08682 rad

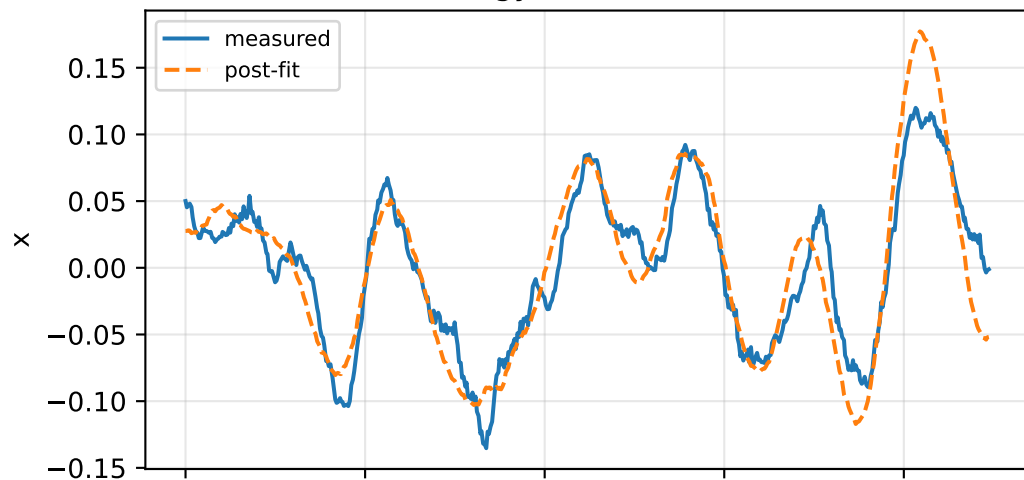


gimbal 4: RMSE=0.02598, max=0.07261 rad

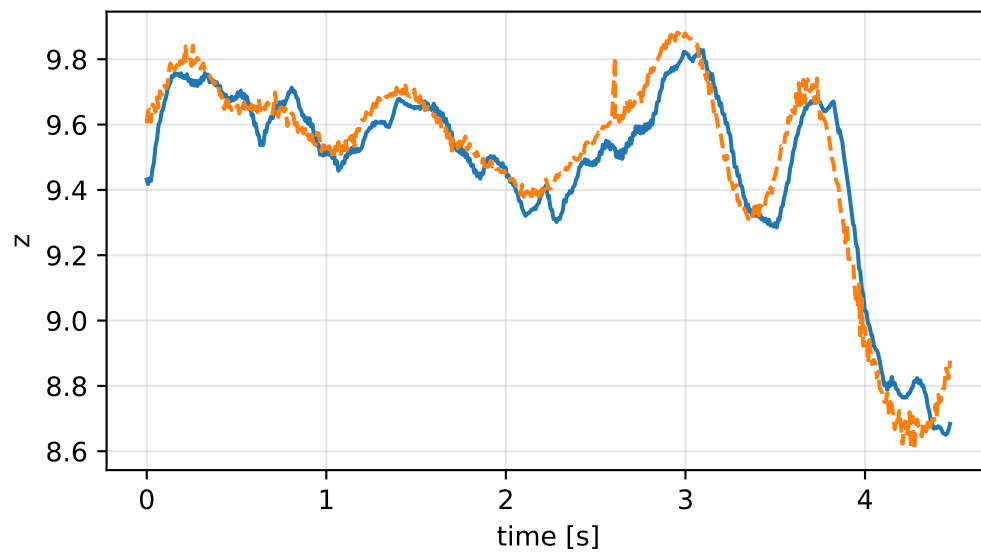
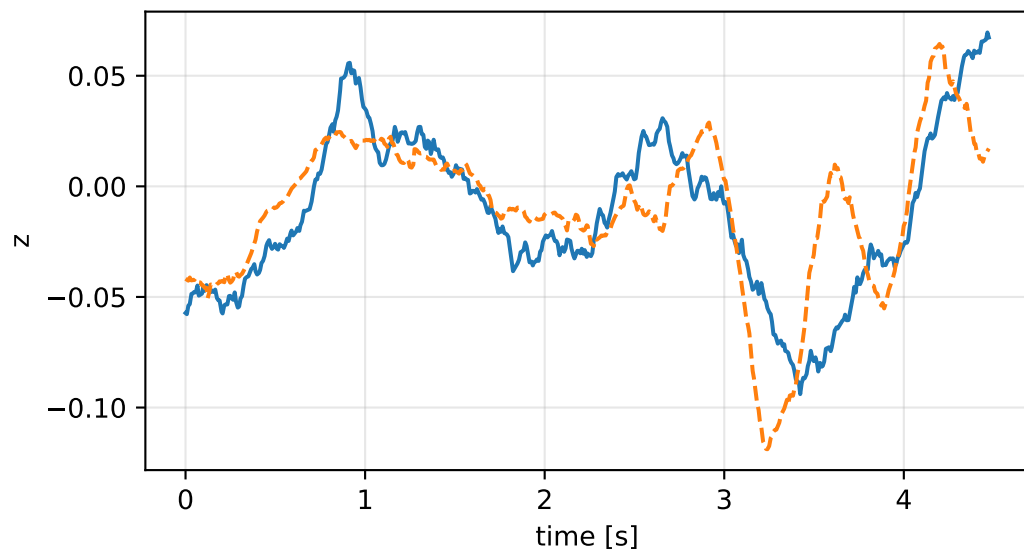
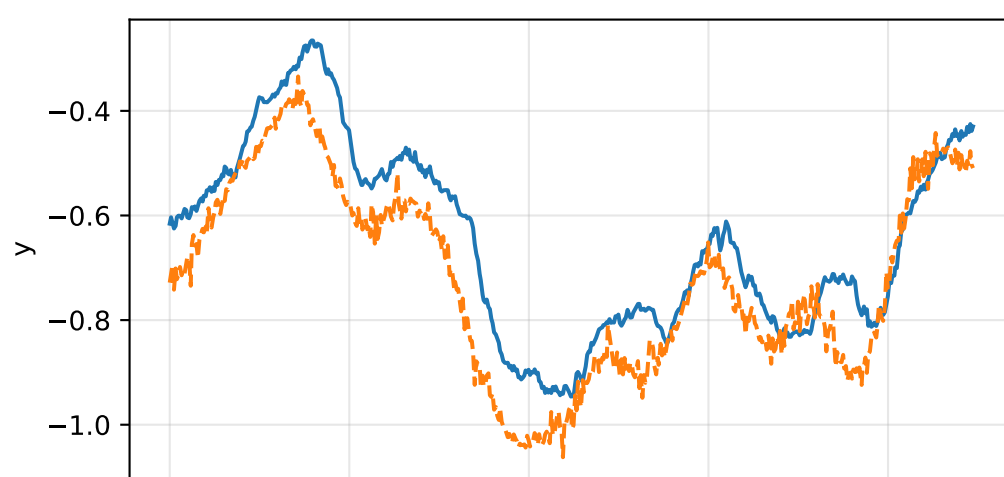
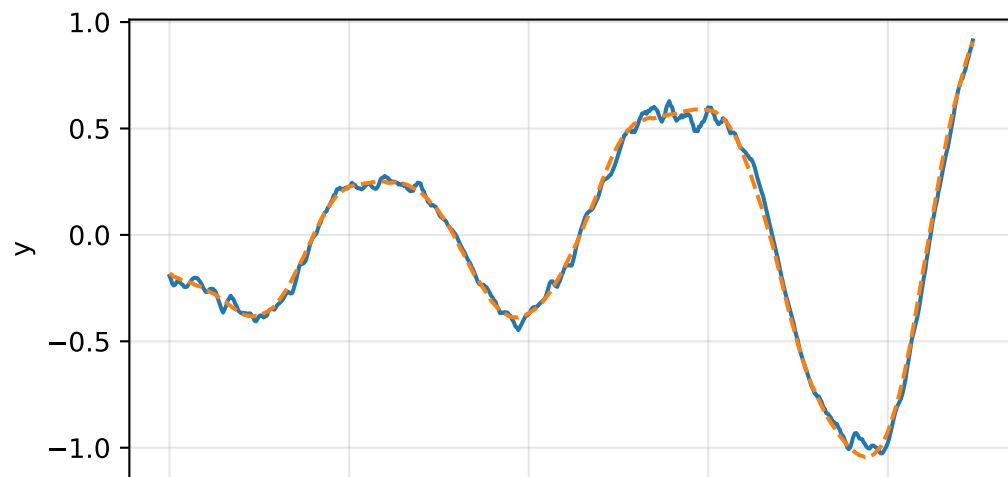
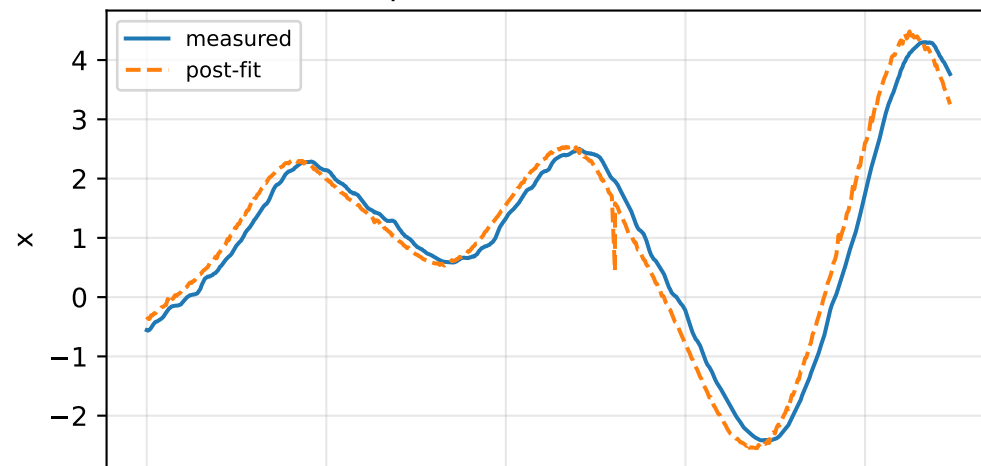


gimbal_measured_linear: IMU comparison (diagnostic only)

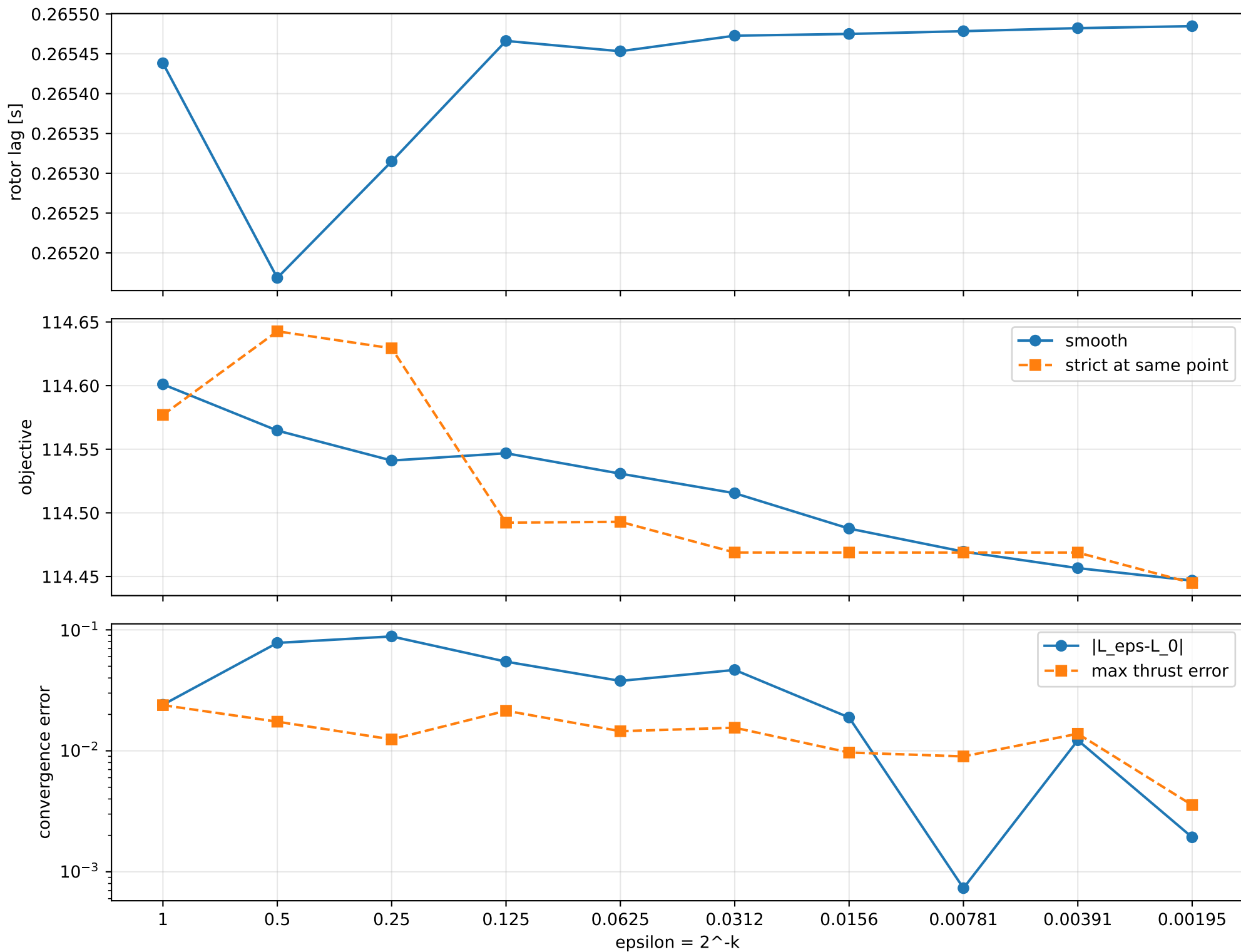
gyro [rad/s]



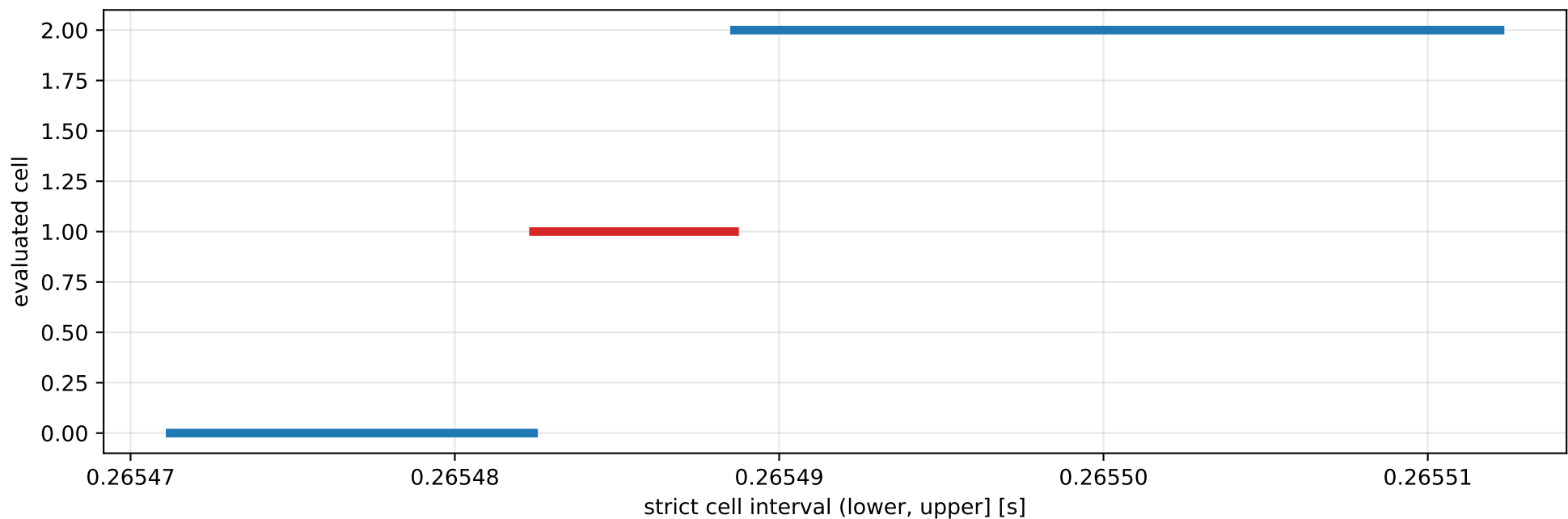
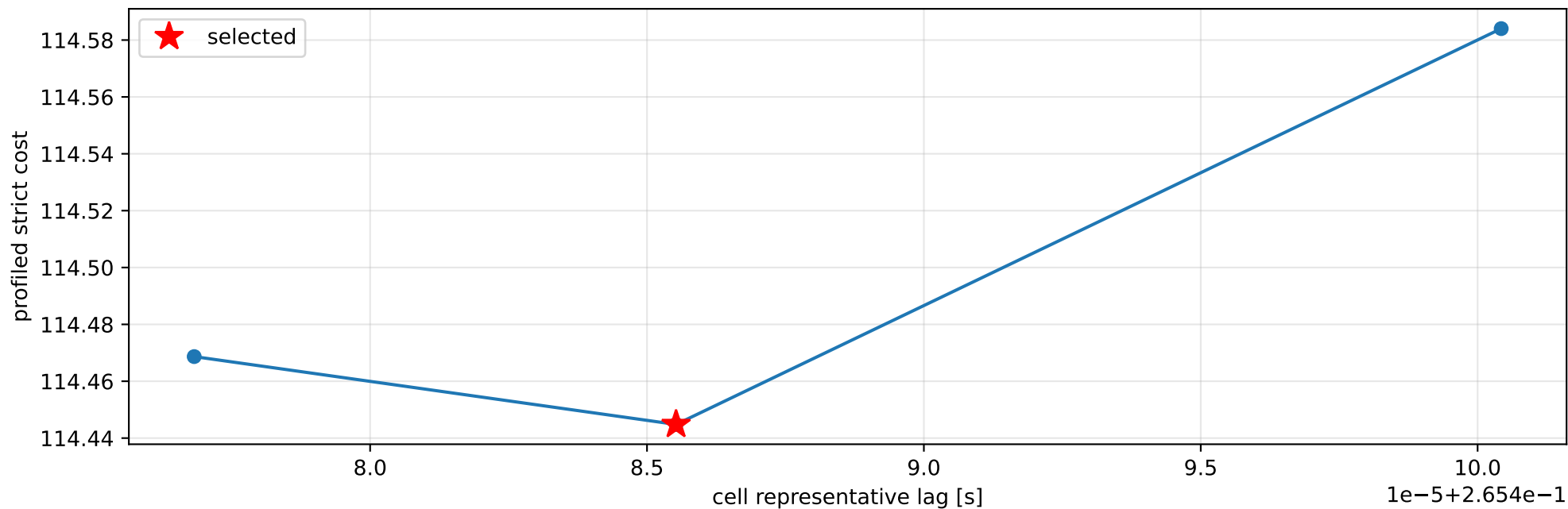
specific force [m/s^2]



gimbal_measured_linear: smooth-to-strict continuation



gimbal_measured_linear: exact strict-ZOH lag cells



selected (0.265482426, 0.265488625] s; final smooth 0.265484674 s; data boundary=False

gimbal_measured_linear: parameter estimate

Case: gimbal_measured_linear

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 4.56930542

inertia [kg m²]:

```
[[ 0.32430549  0.00817182  0.0031829 ]  
 [ 0.00817182  0.17414835 -0.05211929]  
 [ 0.0031829  -0.05211929  0.48119756]]
```

CoG body [m]: [-0.02653459 0.01201455 0.00610655]

force effectiveness: [1.74099241 1.40766673 1.56336836 1.87189445]

scale-free J/m [m²]:

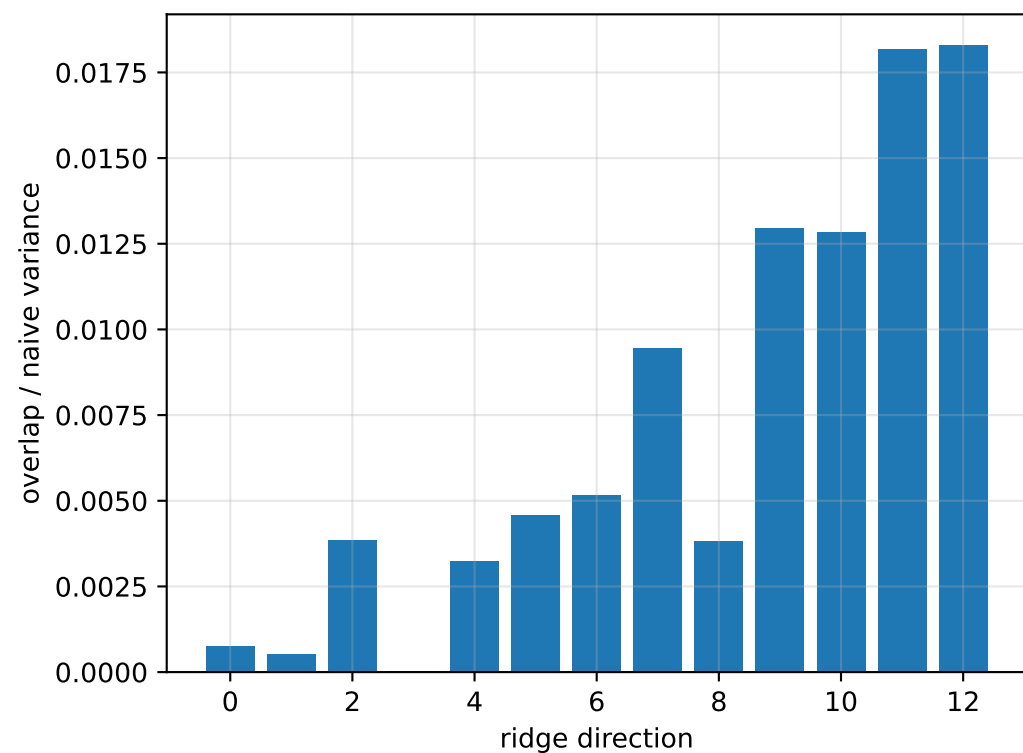
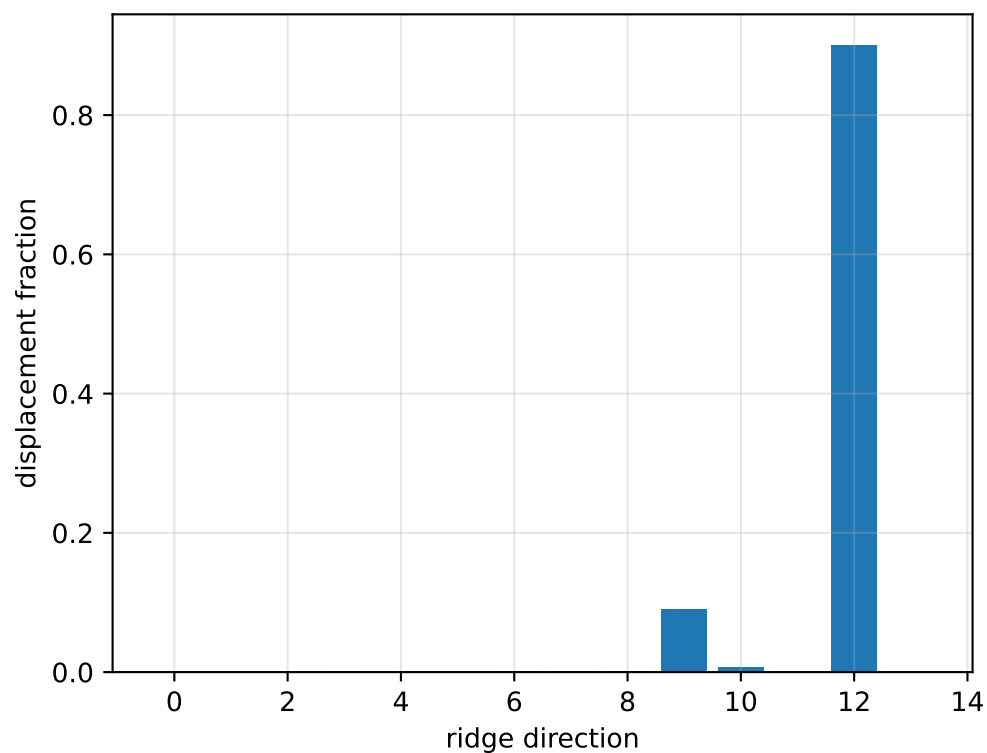
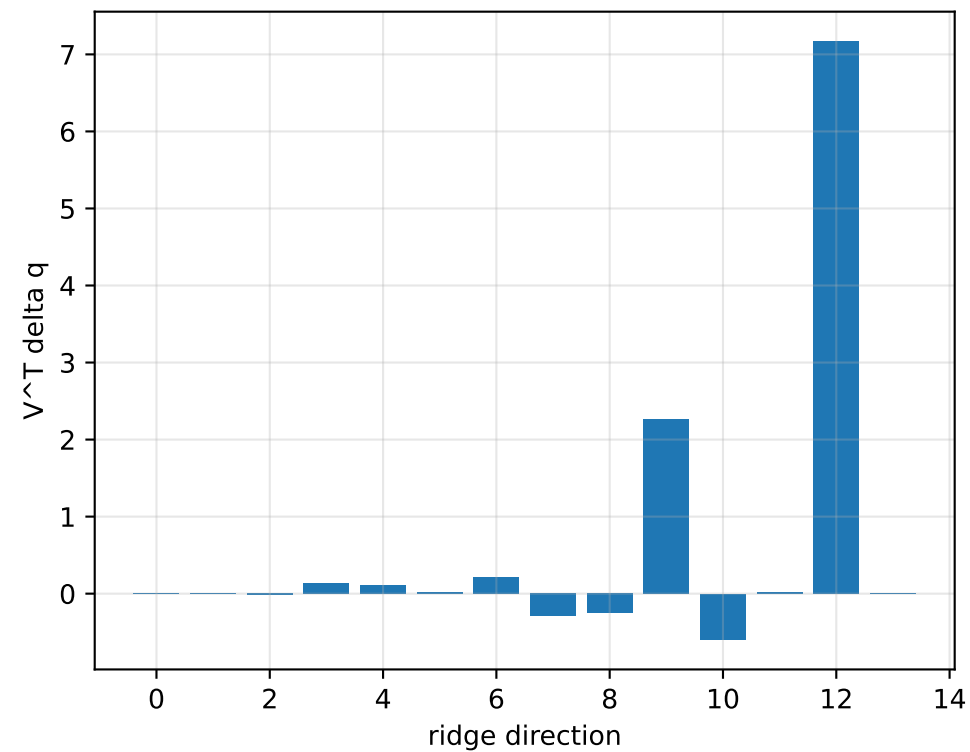
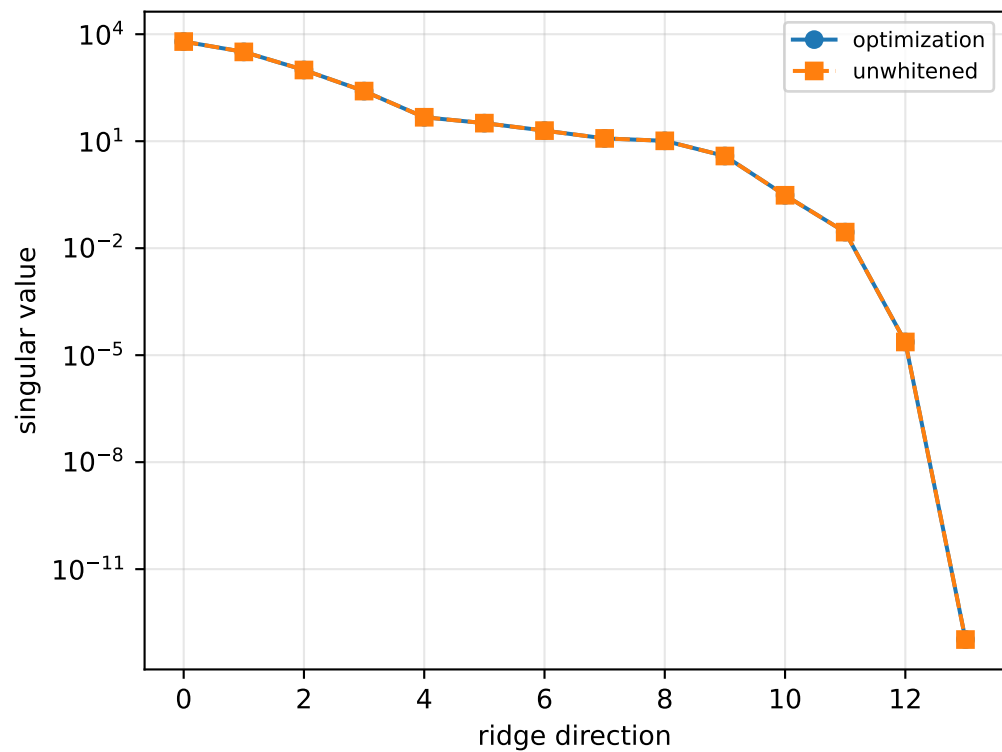
```
[[ 0.07097479  0.00178842  0.00069658]  
 [ 0.00178842  0.03811265 -0.01140639]  
 [ 0.00069658 -0.01140639  0.10531088]]
```

scale-free f/m: [0.38101905 0.30807018 0.34214573 0.40966718]

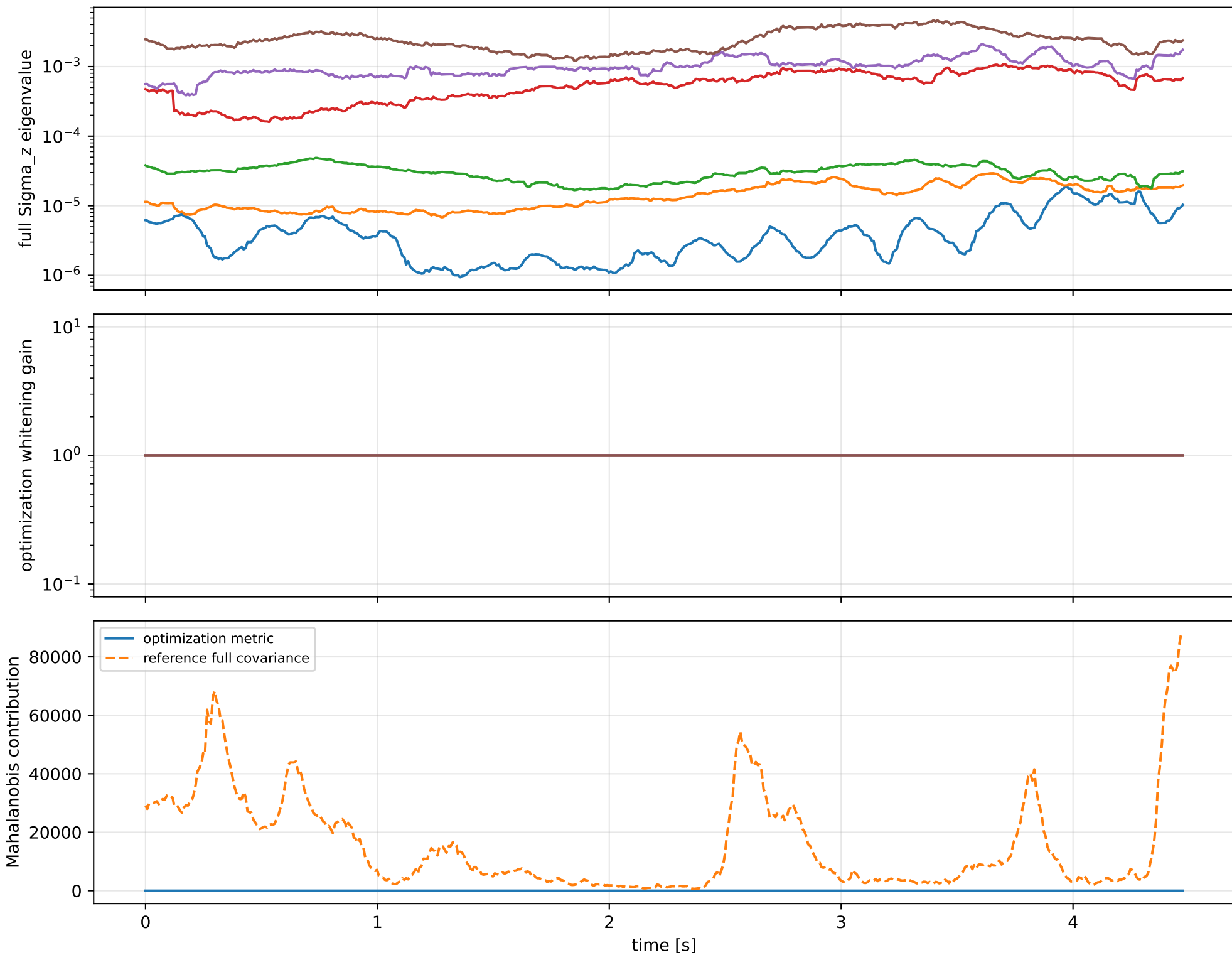
rotor lag cell: (0.265482426, 0.265488625] s

representative: 0.265485525 s

gimbal_measured_linear: ridge spectrum (rank 13, nullity 1, $||\mathbf{J}\mathbf{v}||=1.4\text{e-}13$)



gimbal_measured_linear: optimization vs reference covariance



gimbal_measured_linear: wrench / closure (force max $1.4\text{e-}14$, torque max $2.1\text{e-}16$)

